

GED Math Worksheet: Tables, Graphs, and Interpreting Data - Solutions [cite: 1]

Name: _____ [cite: 2]

Date: _____ [cite: 3]

Instructions: Answer all 30 questions. Show your work where applicable. You may use a calculator for these problems. [cite: 4]

Part A: Tables (Questions 1-6) [cite: 5]

Questions 1-3 refer to the following table: [cite: 6]

Month	Jan	Feb	Mar	Apr
Sales (\$1000s)	45	52	38	61

[cite: 7]

1. **What were the total sales for the four months?** [cite: 8]

Solution: Total Sales = $45 + 52 + 38 + 61 = 196$.

Since the values are in \$1000s, the total is **\$196,000**.

2. **What is the average monthly sales for this period?** [cite: 9]

Solution: Average = $\frac{196}{4} = 49$.

The average monthly sales is **\$49,000**.

3. **By how much did sales increase from February to April?** [cite: 10]

Solution: April Sales (\$61,000) - February Sales (\$52,000) = 9,000.

Sales increased by **\$9,000**.

Questions 4-6 refer to the following table: [cite: 11]

Product	Price	Quantity Sold	Revenue
Widget A	\$8	125	\$1,000
Widget B	\$12	87	\$1,044
Widget C	\$15	92	\$1,380
Widget D	\$10	156	\$1,560

[cite: 13]

4. **Which product generated the highest revenue?** [cite: 14]
Solution: Comparing revenues: Widget A (\$1,000), Widget B (\$1,044), Widget C (\$1,380), Widget D (\$1,560).
Widget D generated the highest revenue.
5. **If Widget B's price increases by 20%, what is the new price?** [cite: 15]
Solution: Current price of Widget B = \$12.
Increase = $\$12 \times 0.20 = \2.40 .
New Price = $\$12 + \$2.40 = \mathbf{\$14.40}$.
6. **What is the total number of widgets sold across all products?** [cite: 16]
Solution: Total = $125 + 87 + 92 + 156 = \mathbf{460}$ widgets.

Part B: Bar Graphs (Questions 7-12) [cite: 17]

Questions 7-9 refer to the following data about monthly temperatures (in °F): [cite: 18]

Month: Jan (32), Feb (35), Mar (45), Apr (58), May (72), Jun (85) [cite: 18]

7. **What is the difference in temperature between the warmest and coldest month listed?** [cite: 19]
Solution: Warmest (June) = 85°F, Coldest (January) = 32°F.
Difference = $85 - 32 = 53^\circ F$.
8. **What is the average temperature for the six months shown?** [cite: 20]
Solution: Average = $\frac{32+35+45+58+72+85}{6} = \frac{327}{6} = 54.5^\circ F$.
9. **Between which two consecutive months is the greatest temperature increase?** [cite: 21]
Solution:
Jan to Feb: $+3^\circ F$
Feb to Mar: $+10^\circ F$
Mar to Apr: $+13^\circ F$
Apr to May: $+14^\circ F$
May to Jun: $+13^\circ F$
The greatest increase ($+14^\circ F$) is between **April and May**.

Questions 10-12 refer to the following data about student enrollment in classes: [cite: 23]

Math (120), Science (95), English (110), History (85), Art (60) [cite: 23]

10. **How many more students are enrolled in Math than in Art?** [cite: 24]
Solution: Math (120) - Art (60) = **60** students.
11. **What is the total enrollment across all five classes?** [cite: 25]
Solution: Total = $120 + 95 + 110 + 85 + 60 = \mathbf{470}$ students.
12. **What percentage of total enrollment is in Science? (Round to the nearest whole percent)** [cite: 26]
Solution: Percentage = $\frac{95}{470} \approx 0.2021$.
Rounded to the nearest whole percent, it is **20%**.

Part C: Line Graphs (Questions 13-17) [cite: 27]

Questions 13-17 refer to the following data showing website traffic (in thousands of visitors) over six weeks: [cite: 28]

Week 1 (12), Week 2 (15), Week 3 (14), Week 4 (18), Week 5 (22), Week 6 (20) [cite: 29]

13. During which week did the website experience the highest traffic? [cite: 30]

Solution: The highest traffic occurred in **Week 5** (22,000 visitors).

14. What was the change in visitors from Week 1 to Week 6? [cite: 31]

Solution: Week 6 (20) - Week 1 (12) = 8.

There was an **8,000 visitor increase**.

15. During which consecutive weeks was there a decrease in visitors? [cite: 32]

Solution: Traffic decreased from **Week 2 to Week 3** (15k to 14k) and from **Week 5 to Week 6** (22k to 20k).

16. What is the average number of visitors across all six weeks? [cite: 34]

Solution: Average = $\frac{12+15+14+18+22+20}{6} = \frac{101}{6} \approx 16.833$.

The average is approximately **16,833 visitors**.

17. If the trend from Week 4 to Week 5 continues, how many visitors would you predict for Week 7? [cite: 35]

Solution: The trend from Week 4 to Week 5 is an increase of 4 (from 18 to 22).

If we apply an increase of +4 to the Week 6 data point (20), the prediction for Week 7 would be $20 + 4 = \mathbf{24}$ (or 24,000 visitors).

Part D: Circle Graphs (Questions 18-22) [cite: 36]

Questions 18-22 refer to the following data about a budget breakdown for a household (\$3,600 dollars total): [cite: 37]

Housing (40%), Food (25%), Transportation (18%), Utilities (12%), Entertainment (5%) [cite: 37]

18. How much money is allocated to Housing? [cite: 38]

Solution: Housing = $\$3,600 \times 0.40 = \mathbf{\$1,440}$.

19. What is the difference between the amount spent on Food and Entertainment? [cite: 39]

Solution:

Food = $\$3,600 \times 0.25 = \900 .

Entertainment = $\$3,600 \times 0.05 = \180 .

Difference = $\$900 - \$180 = \mathbf{\$720}$.

20. If the Entertainment budget is increased by 3%, how much money would be allocated to it? [cite: 40]

Solution: Assuming the budget percentage increases by 3 percentage points (to 8%):

New allocation = $\$3,600 \times 0.08 = \mathbf{\$288}$.

(Note: If interpreted as a 3% increase on the \$180 amount, it would be $\$180 \times 1.03 = \185.40 , but standard GED phrasing implies adjusting the percent share).

21. What percentage of the budget is spent on Housing and Transportation combined? [cite: 41]

Solution: 40% (Housing) + 18% (Transportation) = **58%**.

22. If total household income increases to \$4,000 but percentages remain the same, how much will be spent on Food? [cite: 42]
Solution: New Food Budget = $\$4,000 \times 0.25 = \$1,000$.

Part E: Scatter Plots (Questions 23-26) [cite: 44]

Questions 23-26 refer to the following data showing the relationship between study hours and test scores: [cite: 45]

Study Hours: 1, 2, 3, 4, 5, 6

Test Scores: 62, 68, 75, 81, 87, 92 [cite: 46]

23. What type of correlation does this data show? [cite: 47]
Solution: As study hours increase, test scores consistently increase. This shows a **strong positive correlation**.
24. Based on the pattern, what would you estimate the test score to be for 7 hours of study? [cite: 48]
Solution: The scores increase by roughly 5 to 7 points per hour.
Estimating from the 6-hour mark (92) with an average increase of +6: $92 + 6 = 98$.
25. What is the range of test scores shown in the data? [cite: 49]
Solution: Highest (92) - Lowest (62) = **30**.
26. Is there an outlier in this dataset? Explain your answer. [cite: 50]
Solution: **No**, there are no outliers. All test score increases are relatively consistent and follow a steady upward trend without any data points deviating significantly.

Part F: Histograms (Questions 27-29) [cite: 51]

Questions 27-29 refer to the following histogram data showing distribution of ages at a community center: [cite: 52]

Age 18-25 (8), Age 26-35 (15), Age 36-45 (12), Age 46-55 (10), Age 56-65 (5) [cite: 53]

27. What is the total number of people represented in the histogram? [cite: 54]
Solution: Total = $8 + 15 + 12 + 10 + 5 = 50$ people.
28. Which age group has the highest frequency? [cite: 55]
Solution: The highest frequency is 15, which is the **26-35** age group.
29. What percentage of people are age 46 or older? (Round to the nearest whole percent) [cite: 57]
Solution: People 46 or older = 10 (from 46-55) + 5 (from 56-65) = 15 .
Percentage = $\frac{15}{50} = 0.30$.
The percentage is **30%**.

Part G: Box Plots (Question 30) [cite: 58]

Question 30 refers to the following dataset: 12, 15, 18, 22, 25, 28, 30, 32, 35 [cite: 59]

30. Find the median, lower quartile, upper quartile, and interquartile range for this dataset. [cite: 59]
Solution: The dataset has 9 ordered numbers.

- **Median (Q2):** The middle number is **25**.
- **Lower Quartile (Q1):** The median of the lower half (12, 15, 18, 22) is $\frac{15+18}{2} = \mathbf{16.5}$.
- **Upper Quartile (Q3):** The median of the upper half (28, 30, 32, 35) is $\frac{30+32}{2} = \mathbf{31}$.
- **Interquartile Range (IQR):** $Q3 - Q1 = 31 - 16.5 = \mathbf{14.5}$.

End of Worksheet [cite: 60]